Lecture 25

Erik An

<2025-11-25 Tue>

Contents

1 Cardinality of Sets **1**
1.1 Finite Sets . . . 1
1.2 Infinite Sets and Equivalence . . . 2
1.2.1 Definition . . . 2
1.2.2 Theorem: Equivalence Relation . . . 2

2 Countable Sets **2**
2.1 Definition . . . 2
2.1.1 Terminology Note . . . 2
2.2 Examples . . . 3
2.3 Theorems on Countability . . . 3
2.3.1 Theorem: Infinite Subsets . . . 3
2.3.2 Theorem: Rationals ($\mathbb{Q}$) . . . 3
2.3.3 Theorem: Union of Countable Sets . . . 4

3 Uncountable Sets **4**
3.1 Comparing Cardinalities . . . 4
3.2 Cantor's Theorem: Uncountability of (0,1) . . . 4
3.3 Theorem: Supersets of Uncountable Sets . . . 5

1 Cardinality of Sets

1.1 Finite Sets

Let A and B be finite sets. We denote $|A|$ as the number of elements in A. Two sets have the same size if they have the same number of elements:

$$|A| = |B| \iff \text{number of elements in } A = \text{number of elements in } B$$

1.2 Infinite Sets and Equivalence

For infinite sets, we cannot simply count elements.

1.2.1 Definition

Two sets A and B are said to have equal size (or **cardinality**), or are **equivalent**, if there exists a bijection from A to B.

- **Notation:** $A \sim B$ or $|A| = |B|$.

1.2.2 Theorem: Equivalence Relation

The relation $\sim$ is an equivalence relation. **Proof:**

1. **Reflexive:** $A \sim A$. The identity map $id_A : A \to A$ is a bijection.
2. **Symmetric:** If $A \sim B$, then $B \sim A$. If $f : A \to B$ is a bijection, then the inverse function $f^{-1} : B \to A$ exists and is also a bijection.
3. **Transitive:** If $A \sim B$ and $B \sim C$, then $A \sim C$. Let $f : A \to B$ and $g : B \to C$ be bijections. The composition $g \circ f : A \to C$ is a bijection.

2 Countable Sets

2.1 Definition

A set A is called **countable** (or denumerable) if $A \sim \mathbb{N}$ (the set of natural numbers).

- **Notation:** The cardinality of $\mathbb{N}$ is denoted by $\aleph_0$ (aleph-null).

2.1.1 Terminology Note

There is sometimes ambiguity in terminology:

1. **Denumerable / Infinitely Countable:** Specifically $A \sim \mathbb{N}$.
2. **Countable:** Either finite or denumerable.

In this context, we often focus on infinite countable sets.

2.2 Examples

1. **Hilbert's Hotel:** An infinite hotel with rooms $1, 2, 3, \ldots$ is full. If a new guest arrives, move guest in room n to $n+1$. Room 1 becomes free. The set $\mathbb{N} \cup \{x\} \sim \mathbb{N}$.

2. **Even Natural Numbers:** Let $E = \{2n : n \in \mathbb{N}\} = \{2, 4, 6, \ldots\}$. Define $f : \mathbb{N} \to E$ by $f(n) = 2n$. This is a bijection, so $E \sim \mathbb{N}$.

3. **Integers ($\mathbb{Z}$):** $\mathbb{Z} = \{0, 1, -1, 2, -2, \ldots\}$. We can list them: $0 \to 1, 1 \to 2, -1 \to 3, 2 \to 4, -2 \to 5, \ldots$ Thus $\mathbb{Z} \sim \mathbb{N}$.

2.3 Theorems on Countability

2.3.1 Theorem: Infinite Subsets

Every infinite subset of $\mathbb{N}$ is countable. **Proof Sketch:** Let $A \subset \mathbb{N}$ be infinite. We can list elements of A in increasing order:

- $a_1 = \min A$
- $a_2 = \min(A \setminus \{a_1\})$
- $a_3 = \min(A \setminus \{a_1, a_2\})$
- $\ldots$

This defines a bijection $f : \mathbb{N} \to A$ where $f(i) = a_i$.

2.3.2 Theorem: Rationals ($\mathbb{Q}$)

The set of rational numbers $\mathbb{Q}$ is countable ($|\mathbb{Q}| = \aleph_0$). **Proof Strategy:**

1. Show positive rationals $\mathbb{Q}_{>0}$ are countable. Arrange fractions p/q in a 2D grid: $$ $$\begin{array}{ccccc} 1/1 & 1/2 & 1/3 & 1/4 & \ldots \\ 2/1 & 2/2 & 2/3 & 2/4 & \ldots \\ 3/1 & 3/2 & 3/3 & 3/4 & \ldots \\ 4/1 & 4/2 & 4/3 & 4/4 & \ldots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{array}$$ $$ Traverse diagonals (snake-like path): $\frac{1}{1}, \frac{1}{2}, \frac{2}{1}, \frac{3}{1}, \frac{2}{2}, \frac{1}{3}, \ldots$. Skip fractions that are not in simplest form (repetitions like $2/2$). This creates a listing $q_1, q_2, q_3, \ldots$.

2. Extend to all rationals $\mathbb{Q}$ by interleaving: $\mathbb{Q} = \{0, q_1, -q_1, q_2, -q_2, \ldots\}$.

2.3.3 Theorem: Union of Countable Sets

The union of countably many countable sets is countable. **Proof Sketch:** Let $A_1, A_2, A_3, \dots$ be countable sets. Let $A_n = \{a_{n1}, a_{n2}, a_{n3}, \dots\}$. We can arrange all elements a_{nm} in a grid and traverse diagonals (similar to the proof for $\mathbb{Q}$).

$$\bigcup_{n=1}^{\infty} A_n = \{a_{11}, a_{21}, a_{12}, a_{13}, a_{22}, a_{31}, \dots\}$$

3 Uncountable Sets

3.1 Comparing Cardinalities

- We say $|A| \leq |B|$ if there exists an **injection** from A to B.
- We say $|A| < |B|$ if $|A| \leq |B|$ and $|A| \neq |B|$ (i.e., no bijection exists).

3.2 Cantor's Theorem: Uncountability of (0,1)

The open interval $(0, 1)$ is uncountable. **Proof (Cantor's Diagonal Argument):** Suppose for contradiction that $A = (0, 1)$ is countable. Then we can list all real numbers in $(0, 1)$ as a sequence $r_1, r_2, r_3, \dots$. We use the unique decimal representation (avoiding infinite strings of 9s).

$$\$\$ \begin{aligned} r_1 &= 0.a_{11}a_{12}a_{13}\dots \\ r_2 &= 0.a_{21}a_{22}a_{23}\dots \\ r_3 &= 0.a_{31}a_{32}a_{33}\dots \\ &\vdots \end{aligned} \$\$$$

We construct a new number $b = 0.b_1b_2b_3\cdots \in (0, 1)$ that is not in the list. Define the digit b_n as:

$$b_n = \begin{cases} 1 & \text{if } a_{nn} \neq 1 \\ 2 & \text{if } a_{nn} = 1 \end{cases}$$

(Note: Using 1 and 2 ensures the decimal expansion does not end in infinite 9s, avoiding representation ambiguity).

For every n, the number b differs from r_n in the \$n\$-th decimal place ($b_n \neq a_{nn}$). Therefore, $b \neq r_n$ for any $n \in \mathbb{N}$. This contradicts the assumption that the list contains all numbers in $(0, 1)$. Thus, $(0, 1)$ is uncountable.

3.3 Theorem: Supersets of Uncountable Sets

Let $A \subset B$. If A is uncountable, then B is uncountable. **Proof:** Assume B is countable. Since any subset of a countable set is countable, A must be countable. This contradicts the hypothesis that A is uncountable. Therefore, B must be uncountable.